

Department of Electrical & Electronic Engineering

Brac University

Fall 2025

Project Report

Course Name: Energy Conversion 1 Laboratory

Course Code: EEE 221L

Group No: 03

Section: 03

18

Project Title: Design, Construction and Testing of a Single-Phase, Tap Changing Transformer.

Submitted By:

Name	ID	Contributing Chapter/s
Moobtasim Tajwar	24121190	Title - 3-5
Abul Azmain Afik	24121200	Title - 8-9
Souvik Barman Redul	24121205	Title - 6-7

Rajesh Wagh	24121191	Title - 1-2
Darshan Sarda	24121196	Title - 10-13

Submitted To:

Date of the submission:

Contents

Title	Page
A Project Abstract	
Introduction	
Theory of Transformer	

Designs and Specifications	
Construction Details	
Experimental Setup	
Testing Procedures & Results	
Equivalent Circuit Parameters	
Efficiency Calculations	
Error Analysis & Discussion	
Budget & Cost Analysis	

Limitations & Troubleshooting

Conclusion

References

Abstract:

This report describes the design-~~spec~~ specification, construction and experimental evaluation of a single-phase tap changing transformer developed for laboratory and educational use. The transformer is rated at 48 VA, operates at a frequency of 50 Hz, and steps down a nominal primary voltage of 220 V AC to a secondary voltage of 24 V AC. To facilitate the study of voltage regulation and supply variation, primary-side tap changing of $\pm 5\%$ was incorporated into the design.

The report presents the theoretical background of transformer operation, key design calculations, construction methodology and detailed experimental testing procedures. Standard transformer test - including polarity test, open circuit test, short-circuit test, load test and tap-voltage verification - were performed. Based on experiment data, transformer losses, efficiency and equivalent circuit parameters were evaluated. The results show good agreement between theoretical predictions and practical observations, demonstrating that the constructed transformer

performs reliably within its rated operating conditions.

Introduction:

Transformers are static electromagnetic devices used to transfer electrical energy from one circuit to another at the same frequency but at different voltage levels through the principle of mutual induction. They are indispensable components in electrical power systems, electronic power supplies, and laboratory equipment due to their high efficiency, reliability, and ability to provide electrical isolation.

In real-world power systems, variations in supply voltage can negatively affect the performance and life span of electrical equipment. To mitigate this issue, tap-changing transformers are employed. By changing the effective number of turns in a winding, tap changers allow controlled adjustment of the output voltage without altering the supply frequency.

The objective of this project is to design and analyse a low-rating single phase tap-changing transformer

suitable for educational & laboratory experiments.

The project aims to provide practical insight into transformer construction, operating principles, voltage regulation, losses, efficiency and performance characteristics. Emphasis is placed on theoretical concepts with experimental results obtained through standard transformer tests.

Theory :

~~2-3~~ Principle of operation: A transformer operates based on the principle of mutual induction. When an alternating voltage is applied to primary winding it causes an alternating current flow, which produce a time-varying magnetic flux in the transformer core. This magnetic flux links both the primary and secondary windings. According to Faraday's law of electromagnetic induction, a changing magnetic flux induces an electromotive force (EMF) in any winding it links.

Induced EMF Equation:

The RMS value of the induced EMF in a transformer is given by:

$$E = 4.44 f B_m A N$$

where, E = RMS induced EMF (V)

f = supply frequency (Hz)

B_m = maximum magnetic flux density (T)

A = effective cross-sectional area of the core (m^2)

N = number of turns of the winding

This equation shows that the induced voltage is directly proportional to frequency, magnetic flux density, core area and number of turns.

Voltage and Turns Ratio:

For an ideal transformer, the ratio of primary to secondary voltage is equal to the ratio of the number of turns in the respective windings:

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}$$

By altering the number of effective turns on the primary winding using tap connections, small changes in output voltage can be achieved, which is the fundamental principle behind tap-changing transformers.

Transformer Losses:

Transformer losses reduce efficiency and are mainly classified into two categories:

i) Core (Iron) Losses:

These include hysteresis loss and eddy current loss, both of which occur in the transformer core. Core losses depend primarily on the applied voltage and frequency and remain nearly constant regardless of load.

ii) Copper Losses:

These losses occur due to resistance of the primary and secondary windings and are proportional to the square of the load current (I^2R). Copper losses increase with load and are zero at no-load condition.

Understanding these losses is essential for analyzing transformer efficiency, voltage regulation and thermal performance.

Design Specifications

1. Electrical Specification

- Rated Apparent Power: 48 VA
- Primary Voltage: 220V AC with $\pm 5\%$ Taps
(209V, 220V, 231V)
- Secondary Voltage: 24V
- Rated Secondary Current: 2A
- Frequency: 50 Hz
- Phase: Single Phase

The VA rating was selected to be 48 VA, based on the laboratory current limits, safe operation of measuring instruments, and budget constraints.

2. Core Selection:

An E-I laminated iron core was selected due to its availability, suitability for low frequency operation. During the procurement, the transformer bobbin and lamination set readily available in the local market had a core dimension of 1.9 cm x 2.7 cm.

which led to an effective cross sectional area of 5.13 cm^2 . Rather than trying for a non standard design, this commercially available core size was used for designing the other specification. Given, the flux density of standard laminated ~~core~~ iron ~~and~~ available in the market is $B_m = 1 - 1.2 \text{ Wb/m}^2 \approx 1.1 \text{ Wb/m}^2$ the resulting parameters were calculated and found to be safe within the limit.

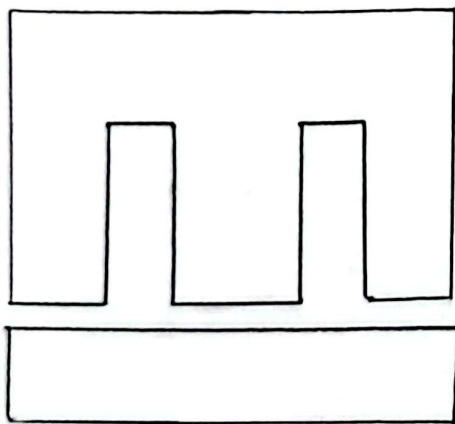


Figure: EI core sheet

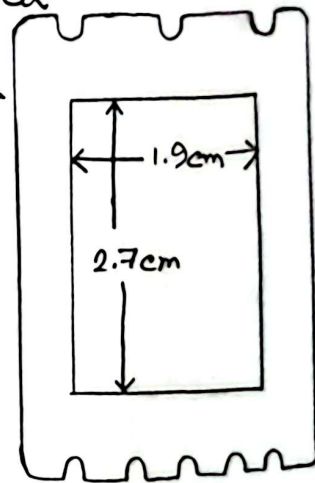


Figure: Bobbin

3. Winding design and Turns Calculation:

The number of turns for the primary and secondary windings was determined using the standard transformer EMF-equation: $E = 4.44 f B_m A N$

$$\rightarrow N = \frac{E}{4.44 f B_m A}$$

where,

Primary voltages

$$E_{P_1} = 209 \text{ V}$$

$$E_{P_2} = 220 \text{ V}$$

$$E_{P_3} = 231 \text{ V}$$

Secondary voltage

$$E_2 = 24 \text{ V}$$

here, E is the EMF generated,
 $f = 50 \text{ Hz}$ frequency, Flux density
 $B_m = 1.1 \text{ Wb/m}^2$ (Standard value
 for laminated sheet)

' A ' is the effective cross sectional
 area

This leads us to select the winding
 turns of

$$f = 50 \text{ Hz}$$

$$B_m = 1.1 \text{ Wb/m}^2$$

$$A = 19 \text{ cm} \times 2.7 \text{ cm}$$

$$= 5.13 \text{ cm}^2$$

$$= 0.000513 \text{ m}^2$$

Primary turns,

$$N_{P_1} = \frac{E_{P_1}}{4.44 \times f B_m A} \approx 1694$$

$$N_{P_2} = \frac{E_{P_2}}{4.44 \times f B_m A} = 1699.81 \approx 1700$$

$$N_{P_3} = \frac{E_{P_3}}{4.44 \times f B_m A} \approx 1784$$

Secondary turns,

$$N_s = \frac{E_2}{4.44 \times f B_m A} = 185.4 \approx 185$$

The primary Taps were taken at the corresponding turn numbers from a single winding to ensure the $\pm 5\%$ voltage variation.

4. Conductor Selection:

The VA Rating of 48 VA allows for maximum 2A on the secondary.

$$\therefore I_p = \frac{V_s I_s}{V_p} = \frac{24V \times 2A}{220V} = 0.22A$$

The max allowed current in the primary is 0.22A.

This led to choose thinner wire for the primary and much thicker wire in the Secondary.

Based on the ~~avail~~ availability in the market as well as within the practical limitation of winding and to limit the I^2R losses the wires selected for

Primary: 38 AWG

Secondary: 24 SWG

Construction details:

The transformer ~~core~~ was fabricated using an EI laminated Core. The primary winding was first wound on the bobbin, followed by inner winding insulation. The secondary winding was then wound over the primary. The tap position for the primary were taken out immediate turns with ~~ins~~ insulated leads. The core ~~was~~ laminations were tightly clamped to reduce air gaps, vibration and noise.

The physical construction was carried out by a vendor based on the provided specifications and designed parameters and ratings were clearly defined and verified through experiments and tests.

5. Experimental Setup, tools, and Instruments:

Instruments:

- AC Voltmeter (Multimeter-digital)
- AC Ammeter
- Wattmeter
- Variable AC Source (0-240V)
- Connecting Leads and terminal blocks

Safety Practices:

- All tests were conducted with proper insulation
- Tap changing was performed with low dc-method when the transformer was de-energized.
- Variable AC source was set to zero before energizing circuits and when not performing tests.

Perform Tests

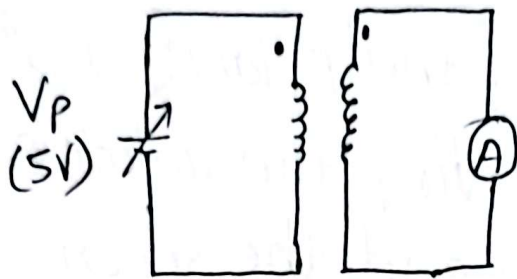
Polarity test

Object: The main object of the polarity test using DC method is to identify the relative Polarity of the Primary and the secondary windings of the transformer. This tests ensure the correct terminals of ~~the~~ transformer. So that the transformer can be safely connected in series or parallel way.

Theory

Polarity of a transformer refers the direction of induced voltage in the secondary winding with respect Primary winding. When DC ~~current~~ current is applied the magnetic flux produced in the core. A momentary voltage is induced in the secondary voltage. The direction of induced voltage depends on the winding orientation. By observing the DC Ammeter we can detect the orientation of the transformer. ~~when~~

circuit diagram



Procedure

- 1) connect DC supply in one side.
- 2) Connect Analog Ammeter in other side.
- 3) Turn on supply for a little moment.

Discussion and verification

When we turn ~~off~~ on the ~~sup~~ supply it heats a direction. If it heat clock-wise direction, it means that the current flows from primary to secondary windage. And if heat the anti-clock-wise direction. Then it verify that the current is flow from secondary to Primary windage. It helps ~~the~~ to detect the primary and secondary windage.

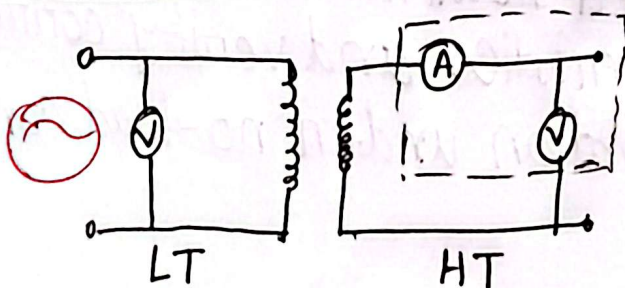
No-load test

Object: The main objective of the no-load test is to determine the no-load current, core loss and no-load PF of the transformer when the secondary winding is open and circuited.

Theory

In the no-load test, rated voltage is applied in Primary winding and secondary winding remain open. For this reason, transformer draw a small current. This current is mainly responsible for the losses. Since the current is very small, so the losses are negligible. Therefore, the power measured during this test approximately equal the core loss of the transformer.

Circuit diagram



Data

$$P_{oc} = 1.6 \text{ W}$$

$$Q = 5.5 \text{ VAR}$$

$$V_{oc} = 24.2 \text{ V}$$

$$PF = \cos \theta$$

$$I_{oc} = 0.274 \text{ A}$$

$$\begin{aligned} &= \cos(73.77) \\ &= 0.27 \end{aligned}$$

$$\begin{aligned} \tan \theta &= \frac{Q}{P} \\ \theta &= \tan^{-1}\left(\frac{Q}{P}\right) \\ &= 73.77 \end{aligned}$$

Procedure

- 1] The secondary winding was kept open.
- 2] Rated AC voltage apply in Primary winding.
- 3] connect voltmeter in primary winding and connect voltmeter, Ammeter and wattmeter in secondary side.
- 4] Take reading when value became stable.

Discussion

The no load current is very small compared to the rated current, which confirms proper core magnetization. The wattmeter reading represents the iron loss and copper loss is negligible at no-load. The obtained ~~result~~ results follow standard transformer characteristics and verify correct transformer operation under no-load condition.

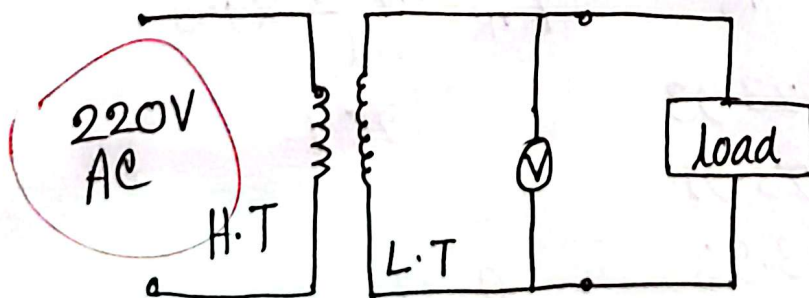
Load test and regulation

Aimed: The objective of the load test is to study the performance of the transformer with load and also determine its voltage regulation.

Theory

When a load is connected to the secondary winding, current flows through the winding. This causes voltage drops in the transformer due to winding resistance and leakage resistance. As a result the voltage decreases with increasing load. Voltage regulation is a measure of this voltage change from no-load to full load and indicates the quality of voltage maintenance by the transformer.

Circuit diagram



Procedure

- 1) Rated voltage was applied to the primary winding.
- 2) load connected to the secondary winding.
- 3) load was increased gradually in steps.
- 4) secondary voltage and current recorded for each load condition.

Data Table

$$V_{nl} = 23.91V$$

Resistive load	V_f	$\%VR = \frac{V_{nl} - V_f}{V_{nl}} \times 100$
2000 Ω	23.1	3.39%
1200 Ω	23.82	0.37%
800 Ω	23.78	0.54%

$$\begin{aligned} V_{nl} &= 23.99 \\ V_{in} &= 222 \end{aligned}$$

Inductive load	V_f	$\%VR = \frac{V_{nl} - V_f}{V_{nl}} \times 100$
7.6 H	23.93	0.25%
3.8 H	23.91	0.33%
2.5 H	23.90	0.37%

capacitive load	V_f	$\% V_R$
1.5	24.05	-0.25%
2.5	24.07	-0.33%
6.5	24.19	-0.83%

Discussion

It was observed that the secondary voltage decreased as the load is increased. This voltage drop is due to internal impedance of the transformer winding. This behavior matches theory, which verifies proper transformer operation under load conditions.

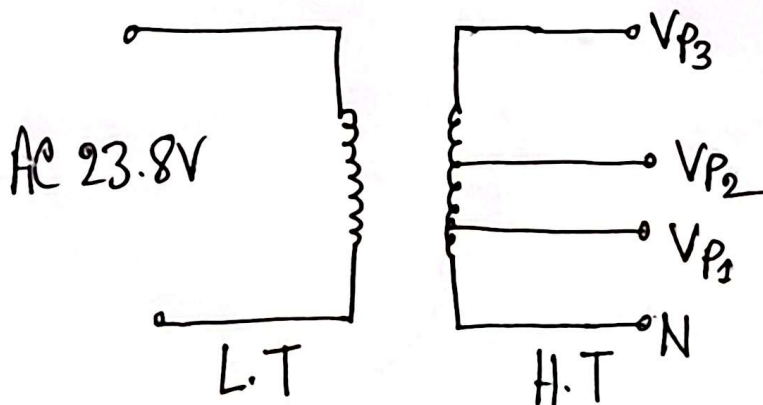
Tap voltage verification.

object: The main objective of the test is to verify the output voltage variation of the transformer at different tap-position.

Theory

Tap changing is used to control the output voltage of a transformer by changing the effective number of turns in the primary winding. Increase the number of turns increases the voltage, while decreasing the turns reduces the voltage. This method help to maintain rated output voltage under varying input conditions.

Circuit diagram



Tap Position
 $V_{P1} = 230.5V$ — +5% Tap — voltage increased
 $V_{P2} = 219.0V$ — Rated Tap — Rated voltage
 $V_{P3} = 208.7V$ — -5% Tap — voltage decreased.

Discussion

In this test, we can observe that when a higher number is selected, the output voltage increases and when the the number of turns reduced, the voltage decreased. The middle tap provides the rated voltage as expected. ~~the~~ All data followed the transformer principal which confirms that the transformer is working properly.

Efficiency

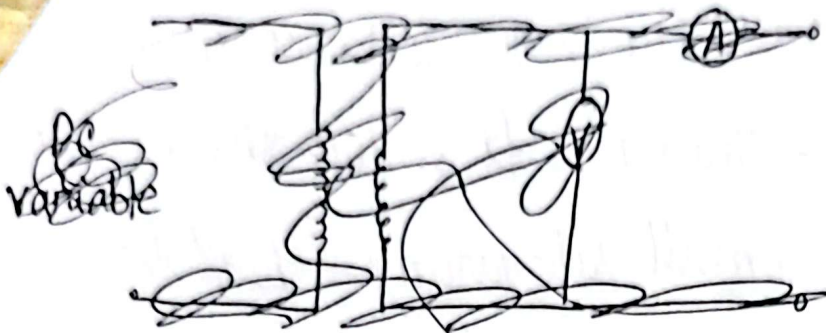
objed: The objective of the efficiency test is to determine the efficiency of the constructed transformer under rated load condition at unity Power factor.

Theory

Efficiency of a transformer is mainly defined the ratio of input and output power. In a transformer losses mainly occur in two form, ~~ie~~ They are cu loss and iron loss. Cu loss depends on load current and iron loss remain constant for a given voltage and frequency. The transformer efficiency is usually high but ~~for~~ some time it became small because of higher losses. It mainly happen for a big error in the transformer. At unity Power factor, the output power is maximum. Mathematically, transformer efficiency is given,

$$\eta = \frac{P_{out}}{P_{out} + P_{cu} + P_{iron}} \times 100\%$$

gait diagram



data

$$P_{out} = 48 \text{ W}$$

$$P_{cu} = 15.2 \text{ W}$$

$$P_{iron} = 1.6 \text{ W}$$

calculation

$$\begin{aligned}\eta &= \frac{P_{out}}{P_{out} + P_{cu} + P_{iron}} \times 100\% \\ &= \frac{48 \text{ W}}{(48 + 15.2 + 1.6) \text{ W}} \times 100\% \\ &= 74.07\%\end{aligned}$$

Discussion

From the calculation, we get the efficiency is 74.07% at unity power factor. The efficiency is lower compared to large commercial transformer, because the constructed transformer has relatively higher cu loss and iron loss .

Since iron loss remains constant and cu loss increases with load, the total loss affects the overall efficiency. This result follow the expected ~~the~~ things for a small laboratory - built transformer and confirms correct transformer operation.

$\therefore \eta =$

8. Error Analyze and Discussion:

□ Theoretical Data and our experimental data have some difference. These values differ due to winding resistance, core losses, measurement tolerances, and supply voltage fluctuations. Percentage error was within acceptable limits for a laboratory built transformer.

9. Budget and Cost analysis:

Our transformer making project cost included the laminated core, copper wire with insulation, laminating sheet, terminals, wires and fabrication charges. Our project was completed within our allocated budget with ~~10~~ 1500/- BDT.

- Transformer bobbin - 30 BDT
 - Laminated Iron - 200 BDT
 - Copper Wire - 480/- BDT
 - Insulating Material - 150/- BDT
 - Transformer outer chasing 100/-
 - Labour Cost = 500 BDT
-
- Total : 1360 BDT

10. Limitations, Issues, and Troubleshooting

1. Design and Calculation Issues:

Determination of core area and turns per volt required several assumptions due to the unavailability of exact standard data for the locally available core.

While calculating tap voltages ($\pm 5\%$), fractional numbers of turns were obtained, which required rounding. This caused a deviation between theoretical and measured voltages.

2. Construction and Material Issues:

The copper wire available in the local market did not perfectly match the calculated wire gauge and insulation

quality. During manual winding, maintaining uniform turns and proper layer insulation was difficult. Slight air gaps were present between laminated core sheets due to imperfect stacking.

3. Measurement troubleshooting:

Instruments were checked for zero errors before each test. Multiple voltage readings were taken and averaged to reduce random measurement error.

11. Conclusion :

In this project, a single-phase tap changing transformer was successfully designed, constructed, and tested according to the given specifications. The design process involved careful calculation of core dimensions, turns per volt, primary and secondary windings and tap positions. Practical construction using laminated iron core and copper windings provided hands-on experience with real transformer components.

Performance tests such as polarity test, no-load test, load test, tap voltage verification, and efficiency calculation

were conducted. The experimental results showed reasonable agreement with theoretical values, with minor deviations caused by material quality, winding imperfections, core losses, and measurement inaccuracies. The tap-changing mechanism effectively provided voltage variation within the specified $\pm 5\%$ range, demonstrating its practical usefulness in voltage regulation.

Although some limitations were faced due to laboratory equipment constraints, time limitations and locally available materials, the overall performance of the transformer was satisfactory for educational purposes. This project enhanced practical understanding

of transformer design, construction, testing procedures, losses voltage regulation and efficiency. It also improved problem-solving skills related to troubleshooting and performance evaluation in real-world electrical systems.

References:

1. EEE 221 L Laboratory Manual, Brunel University.

~~2. Principles of Electrical Machines, V.K. Mehta;~~
~~Rohit Mehta~~

Transformer Lab Test Data

Design and Calculation:

$$VA \text{ Rating} = 48 VA$$

- Primary - 209 - 220 (Nominal) - 231 V
- Secondary - 24 V

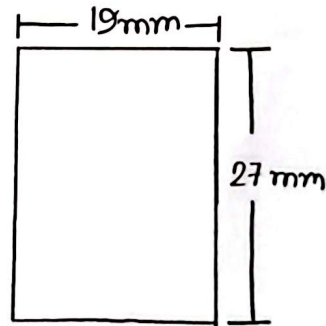
Wire Gauge Selection:

$$\text{Secondary} - 2A (\text{Max}) - 28 \text{ AWG}$$

$$\text{Primary} - I_1 = \frac{V_2 I_2}{V_1} = 0.21 A - 18 \text{ AWG}$$

Core Area:

$$A = (19 \times 27) \text{ mm}^2$$
$$= 0.000513 \text{ m}^2$$



Turns Ratio (From theory)

$$\frac{N_1}{N_2} = \frac{V_1}{V_2} = \frac{220}{24} = 9.167$$

* Turns per Volt:

$$\frac{N_1}{E_1} = \frac{1}{4.44 f \phi} = \frac{1}{4.44 f \cdot B_m A_{\text{core}}}$$
$$= 7.72$$

$$[B_m = 1.1 \text{ T Wb/m}^2]$$

$$N_1 = \frac{E_1}{4.44 \times f \cdot \phi}$$

$$= 1699.81$$

$$\approx 1700$$

$$N_2 = \frac{E_2}{4.44 \times f \times \phi}$$

$$= 185.4$$

$$\approx 185$$

Tap position 5% less

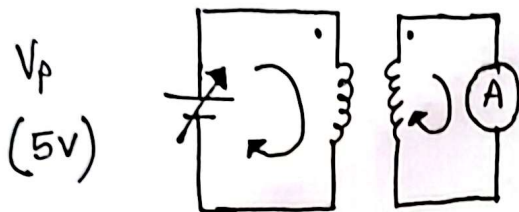
$$N_{P_1} (209V) = 1614$$

Tap position 5% more

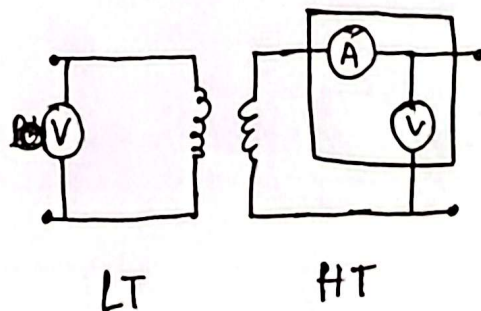
$$N_{P_2} (231V) = 1784$$

Polarity test: (DC Method)

From this test we will confirm the current flow direction or the polarity of the terminals of the transformers.



No load test:



$$P_{oc} = 1.6 \text{ W}$$

$$Q = 5.5 \text{ VAR}$$

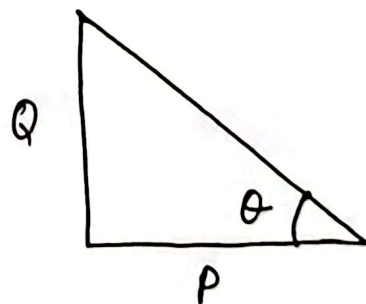
$$V_{oc} = 24.2 \text{ V}$$

$$I_{oc} = 0.274 \text{ A}$$

$$PF = \cos \theta$$

$$= \cos 73.77$$

$$= 0.27$$

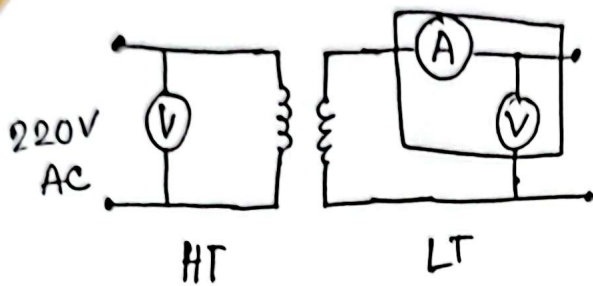


$$\tan \theta = \frac{Q}{P}$$

$$\Rightarrow \theta^\circ = \tan^{-1}\left(\frac{Q}{P}\right)$$

$$= 73.77$$

Short-Circuit Test :



$$P_{sc} = 15.2W$$

$$P_f = \cos \theta = 0.992$$

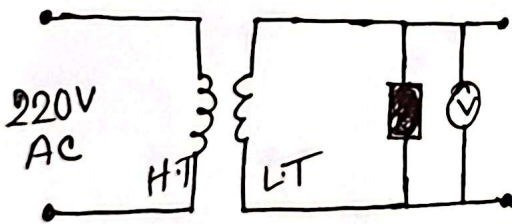
$$V = 72.9V$$

$$\therefore Q = 0.9 \text{ VAR}$$

$$I = 0.21A$$

$$I_{sc} = 1.97A$$

Load-test & Voltage Regulation :



$$V_{nl} = 23.91V$$

Resistive load -	V_p	Regulation %VR = $\frac{V_{nl} - V_p}{V_{nl}} \times 100$
2000 Ω	23.1	3.39%
1200 Ω	23.82	0.37%
800 Ω	23.78	0.54%

Inductive Load	V_f	%VR =
7.64	23.93	0.25%
3.84	23.91	0.33%
2.54	23.90	0.37%

$$V_{nl} = 23.99$$

$$V_{in} = 222$$

Capacitive Load	V_f	%VR
1.5	24.05	- 0.00 0.25%
2.5	24.07	- 0.33%
6.5	24.19	- 0.83%

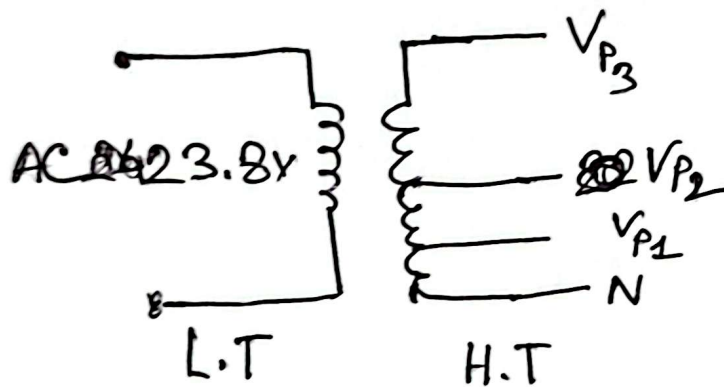
Shahi
28/12/25

Efficiency:

At unity power-factor the output of the transformer will output 48 W

$$\begin{aligned}\therefore \text{efficiency } \eta &= \frac{P_{out}}{P_{out} + P_{cu} + P_{iron}} \times 100\% \\ &= \frac{48 \text{ W}}{48 \text{ W} + 15.2 + 1.6 \text{ W}} \times 100\% \\ &= ~~69.8\%~~ \\ &= 74.07\%\end{aligned}$$

TAP - Verification:



$$V_{P1} = 230.5 \text{ V}$$

$$V_{P2} = 219.0 \text{ V}$$

$$V_{P3} = 208.7 \text{ V}$$